

Xinbao Qiao

✉ xinbaoqiao@cuhk.edu.hk | 🌐 xinbaopedia.top | ● github.com/XinbaoQiao | ★ [Google Scholar](#)

Education

The Chinese University of Hong Kong Ph.D. student in Information Engineering. Advised by *Hong Kong. 2026 - present*
[Angela Yingjun Zhang](#).

Research Areas Data-centric ML and AI with networks, including AI for Networks and Networks for AI.

Zhejiang University M.Sc. in Artificial Intelligence. Advised by [Meng Zhang](#).

Zhejiang. Sep 2022 ~ Dec 2025

GPA and Coursework Major GPA 90/100 (rank 3/25). AI Algorithms and Systems (100); Secure Artificial Intelligence (97).

Shandong. Sep 2018 ~ Jul 2022

Shandong University B.Eng. in Communication Engineering.

GPA and Award Major GPA 82.47/100. Third-class Academic Award, 2018-2019.

Experience

Doctoral Research @ The Chinese University of Hong Kong

Hong Kong. 2026 - present

- Study data-centric ML and AI-network co-design where data, computation, and evaluation signals are distributed across clients, institutions, or networked infrastructure.
- Investigate distributed Wasserstein barycenter methods for constructing shared distributional references without pooling raw data.

Research on Data-Centric ML Systems @ Zhejiang University

M.Eng. Student. Mar 2023 ~ Dec 2025

- Developed theoretically grounded approaches for data influence attribution and certified machine unlearning.
- Studied data-centric trade-offs across fairness, robustness, privacy, and utility in trustworthy AI systems.
- Architected cloud-edge collaborative human-space healthcare frameworks for real-time multimodal monitoring.

Research on Trustworthy LLM Systems @ NUSRI CQ

Full-time Research Intern. Jun 2025 ~ Dec 2025

- Examined illusory pattern perception in LLMs and proposed an interpretable framework for spurious inference behavior.
- Developed Wasserstein-based methods for synthetic-data evaluation under limited or fragmented real-data access.

Selected Publications

Accepted and published papers are listed before under-review manuscripts.

1. **When Sample Selection Bias Precipitates Model Collapse** *ICML, 2026.*
Xinbao Qiao, Xianglong Du, Wei Liu, Jingqi Zhang, Peihua Mai, Meng Zhang, Yan Pang
[Low-resource verification can turn local sample selection into persistent tail pruning.](#)
2. **Beyond Binary Erasure: Soft-Weighted Unlearning for Fairness and Robustness** *AAAI, 2026.*
Xinbao Qiao, Ningning Ding, Yushi Cheng, Meng Zhang
[Soft-weighted corrective unlearning for non-binary fairness and robustness interventions.](#)
3. **Hessian-Free Online Certified Unlearning** *ICLR, 2025.*
Xinbao Qiao, Meng Zhang, Ming Tang, Ermin Wei
[Certified unlearning without explicit Hessian inversion through recollected statistics.](#)
4. **DynFrS: An Efficient Framework for Machine Unlearning in Random Forest** *ICLR, 2025.*
Shurong Wang, Zhuoyang Shen, Xinbao Qiao, Tongning Zhang, Meng Zhang
[Exact and efficient random-forest unlearning in dynamic environments.](#)
5. **Federated Learning as Optimal Transport: Barycentric Multi-Prototype Classification** *Under review.*
Xinbao Qiao, Wenjing Yan, Ying-Jun Angela Zhang
[Barycentric multi-prototype classification for federated representation geometry.](#)
6. **Illusory Pattern Perception Drives Spurious Inference in Large Language Models** *Under review.*
Peihua Mai, Zhuoyan Shao, Xinbao Qiao, Meng Zhang, Xinyue Zhou, Yan Pang
[Analysis of spurious pattern perception and interpretable behavior in LLMs.](#)